Evaluating RAG-Enhanced Transformer Robustness for Toxic Comment Detection under Noisy Text

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1.Introduction

Toxic comment detection is a fundamental task in natural language processing, aiming to automatically identify aggressive, hateful or harmful language in user-generated content. Although Transformer-based models such as BERT have made significant progress, their robustness in the face of noisy text remains limited. This project evaluates the integration of the retrieval enhancement mechanism(RAG) into the Transformer backbone for toxic comment detection in noisy scenarios. The core issue is: Can RAG enhance the robustness of the Transformer in noisy environments?

For this purpose, we compared three methods: Logistic Regression baseline, fine-tuned BERT, and RAG enhanced fine-tuned BERT, and evaluated them in terms of performance and anti-noise stability. The experiment conducted controlled comparisons and hyperparameter adjustments using a nested cross-validation framework to examine whether retrieval enhancement brings measurable benefits compared to pure transformer fine-tuning. The dataset is derived from the Wikipedia reviews of the Jigsaw Toxic Review Classification Challenge(https://www.kaggle.com/c/jigsaw-toxic-comment-classification-challenge). The original 6 types of toxicity were mapped to a binary classification of toxic and non-toxic, in conjunction with a standard preprocessing procedure. To evaluate robustness, we applied controlled perturbations to the test set under different noise intensities, generated corresponding noise text versions, and conducted comparative verifications based on these.

2. Literature Review

Early toxic comment detection approaches mostly relied on traditional methods such as Logistic Regression, but they were insufficient in capturing complex semantics and subtle contexts, especially in identifying disguised or implicit harmful expressions. To address this challenge, Transformer architectures (such as BERT) have significantly improved performance by pre-training generic language representations on massive unlabeled corpora and fine-tuning on downstream classification tasks (Devlin et al., 2018). However, research indicates that introducing external knowledge is the key to improving latent toxicity detection (Lee et al., 2024). To this end, the academic community has explored the combination of retrieval mechanisms with large language models: integrating pre-trained Transformers with kNN modules to retrieve similar contexts from large-scale corpora and fuse them with model outputs can achieve significant gains when dealing with rare patterns and factual knowledge (Khandelwal et al., 2020). Subsequently, Lewis et al. (2020) formally proposed the RAG framework. By providing relevant background for classification models such as BERT, the models can more easily grasp the intent of the text, especially in the determination of ambiguous or harmful content that requires specific knowledge. Based on this, this study integrates RAG into BERT to construct the BERT+RAG post fusion model, verifying whether external retrieval of knowledge can significantly enhance the robustness and overall performance of Transformer in toxic comment detection.

3. Methods

3.1 Data Preprocessing

Jigsaw-toxic-comment-classification-challenge dataset undergoes a comprehensive cleaning pipeline to ensure data quality. To start with, we validate ID consistency by enforcing 16-character hexadecimal format and remove misaligned rows where text is labeled. Comment text cleaning also is able to remove timestamps, excel date numbers, repeated characters (>10 consecutive), and garbled content with excessive unprintable characters (>5% unprintable ratio or >20% replacement characters). Empty comments and entries shorter than 3 characters were excluded. We created a binary toxicity label (is_toxic) by aggregating six original label categories (toxic, severe_toxic, obscene, threat, insult, identity_hate), yielding 151,443 valid samples (10.3% toxic, 89.7% non-toxic). For computational efficiency, we sample 30,000 instances while preserving the original dataset distribution.

The final dataset is partitioned using 10-fold stratified cross-validation with a fixed random seed (42) to ensure reproducibility across all experiments. Besides, To imitate real-world text corruption, we design four types of character- and word-level noise: Character typo: Random character replacement with symbols (*@#$%&); Character swap: Transposition of adjacent characters; Character deletion: Random character removal; Word swap: Transposition of adjacent words. For each validation fold, noisy variants are generated at three intensities (p = 5%, 10%, 20%), representing the proportion of words affected. Noise generation class uses fold-specific random seeds (SEED + fold_id) for reproducibility. Performance degradation (ΔF1 = F1_clean - F1_noisy) quantifies robustness: lower degradation indicates superior resilience to input perturbations.

3.2 Algorithms:

Following the instruction in the project file, we select three algorithms with different model class complexity: Our simple model is Logistic Regression classifier, which is fundamentally linear baseline with almost 10K parameters and $O(n \cdot d \cdot iter)$ training complexity; BERT (medium) represents deep learning capacity with 110M parameters, time complexity is $O(n \cdot L^2 \cdot h \cdot l \cdot epochs)$, and substantial computational cost (~45 min/fold on A100 GPUs); BERT-RAG model (Complex) is the most sophisticated classifier, combining BERT's full 110M-parameter architecture with an additional retrieval system—requiring SentenceTransformer

encoding, FAISS index construction ($O(n \cdot d_emb \cdot \log n)$), and K-nearest neighbor search at inference, resulting in $O(L^2 \cdot h \cdot l) + O(K \cdot d_emb)$ per-sample prediction complexity and maintaining a 27K-sample embedding index in memory.

3.2.1 Logistic Regression (LR) model:
Feature Engineering: We use traditional TF-IDF vectorization with 10,000 maximum features, document frequency thresholds (min_df=5, max_df=0.7), and 1–2-character n-grams. Hyperparameter Tuning: Using nested 3-fold cross-validation from the outer dataset fold, we perform grid search in these parameter sets: regularization strength $C \in \{0.01, 0.1, 0.5, 1.0, 10.0, 100.0\}$, regularization method: L1, L2, none. The hyperparameter maximizing validation macro-F1 is selected for final evaluation on the outer test fold.

3.2.2 BERT Model:
Feature Construction: Input texts are tokenized using the pre-trained BERT-base-uncased tokenizer with truncation at 256 tokens. Hyperparameter Tuning: There are three parameters in fine-tuning for inner 3-fold cross-validation from the outer fold dataset. Grid search is executed in: learning rate $\in \{3\times10^{-5}, 5\times10^{-5}, 7\times10^{-5}\}$, batch size = 16, and epochs = 2. The combination yielding highest validation macro-F1 is applied to train the final model on the full outer training set.

3.2.3 BERT-RAG Post Fusion Model:
Feature Construction: In this model, we employ SentenceTransformer (all-MiniLM-L6-v2) to encode comments into 384-dimensional dense embeddings normalized to unit length. Hyperparameter Tuning: The fusion model requires four hyperparameters: interpolation weight $\alpha \in \{0.0, 0.2, 0.5, 0.8, 1.0\}$, number of neighbors K = 32, temperature $T \in \{0.05, 0.10, 0.15\}$, and classification threshold $\tau = 0.5$. We also perform grid search on clean validation data from the inner 3-fold splits, selecting the configuration that maximizes macro-F1.

4. Results and Discussion:
Nested cross-validation was conducted on 30,000 extremely imbalanced data points, with 10 folds on the outer layer and 3 folds on the inner layer. All evaluation indicators are implemented independently, and the errors are given on the outer folding layer. Under clean data, the Macro-F1 of Fusion model is 0.905±0.013, which is higher than that of BERT (0.893±0.011) and LR (0.850±0.016), as shown in Figure 1, indicating that retrieval enhancement can bring stable small gains under noise-free conditions.
When the noise intensity rose to 0.20, the advantage did not expand significantly. The ΔF1 of Fusion model was 0.047±0.048, and that of BERT was 0.053±0.055. The difference was not significant, as shown in Figure 2. The results that are more sensitive to imbalance are indicated in the same direction: at each noise level, the ΔAUPR of Fusion minus BERT is negative, approximately -0.015 to -0.021, as shown in Figure 7. The robustness slope of F1 fitted by noise intensity is similar, approximately -0.23 to -0.27, as shown in Figure 6. The pairing comparison shows that in 8 out of 10 outer folds, the degradation of Fusion model is no greater than that of BERT, indicating a consistent but moderate improvement. In terms of the minority class, the recall of Fusion in the clean set is approximately 0.84, but it basically disappears under high noise, as shown in Figure 4 and Figure 7.

Figure1

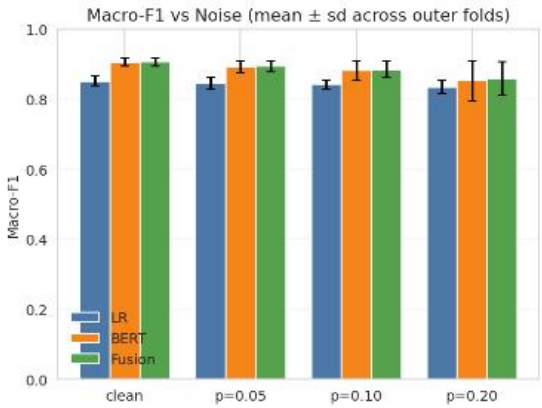


Figure2

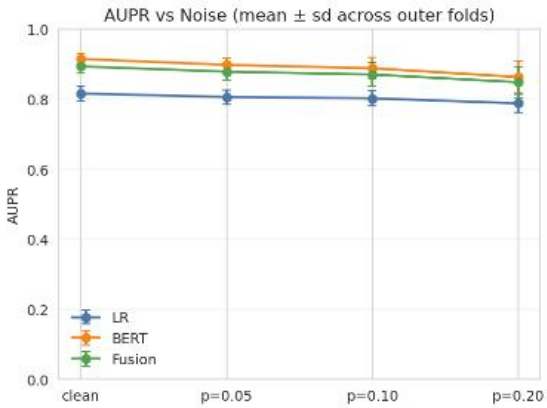


Figure3

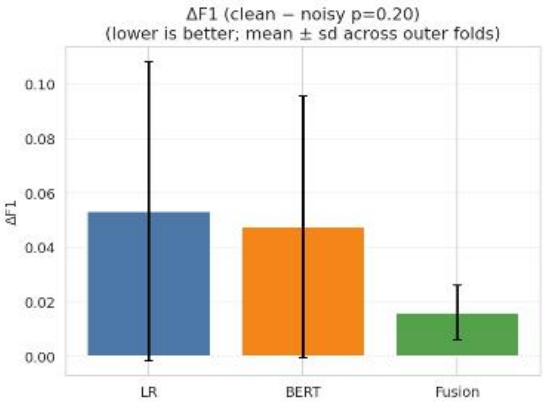
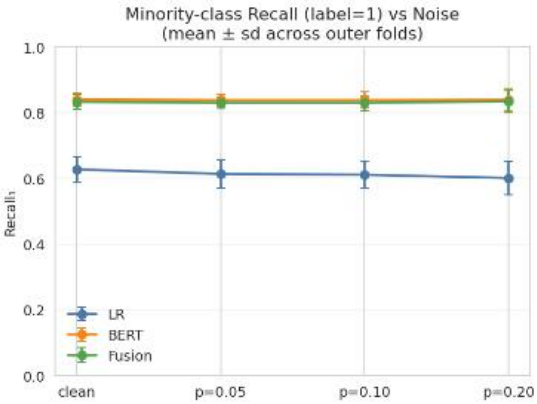


Figure4



The above phenomenon may be explained by the combined effect of the retrieval link and the data distribution. Character-level noise can bias query embeddings, leading to a decline in the similarity of nearest neighbors and making it more difficult to retrieve evidence from minority classes. Under a long-tail distribution of 9:1, this decline is magnified and ultimately manifests as a negative ΔAUPR. The post-fusion implementation adopts a fixed K equal to 32 and static fusion weights and temperatures. When the overall similarity is low, low-quality retrieval still enters the decision-making level, raising the variance. The cross-fold standard deviation of Fusion is approximately 0.025 and the coefficient of variation is about 0.028, which lies between LR and BERT, as shown in Figure 5. It can be understood from this that Fusion benefits from the complementarity of parameterized and non-parameterized information in clean sets. In cases of extreme imbalance and superimposed high noise, the retrieval of evidence becomes thin and biased, and the gain is hedged.

Figure5

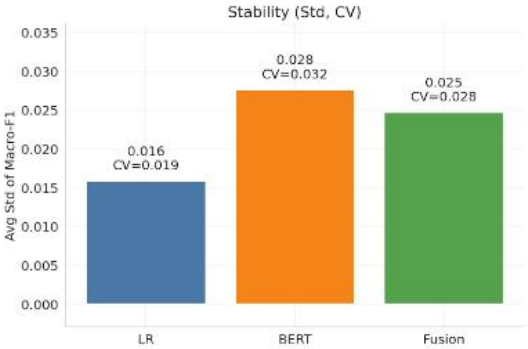
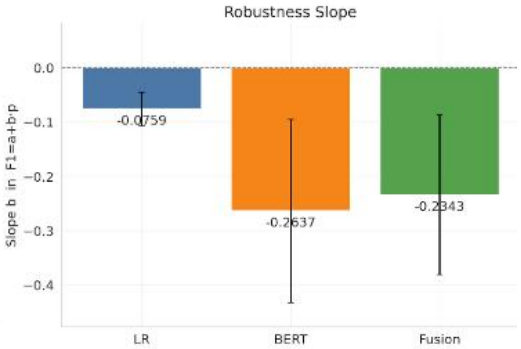


Figure6



All metrics computed from scratch; error bars = 95% CI across 10 outer CV folds.

Figure7

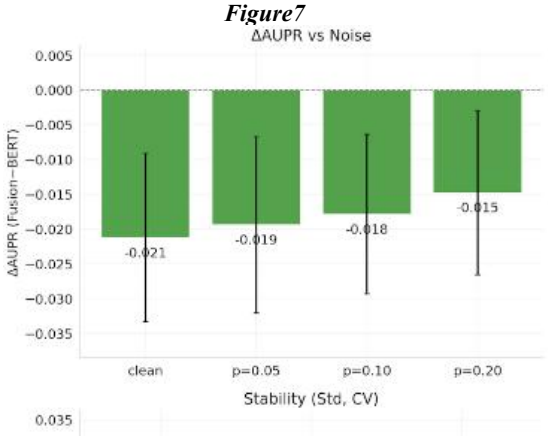
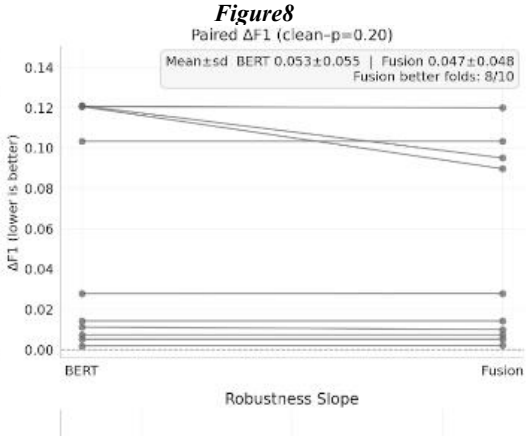


Figure8



Hardware and time budgets also impose constraints on design trade-offs. The cost of nested cross-validation is relatively high. The search space is tightened within the acceptable video memory and duration, fixed K at 32, and did not use cross-encoder rearrangement and sample adaptive gate control, nor did we conduct multiple random seed repetitions and larger-scale confidence interval estimations. To ensure the efficiency of reproduction and visualization, vector retrieval adopts offline caching and batch

evaluation. These strategies ensure that the experiments can be implemented, but they limit the search for the optimal configuration and the verification of significance in high-noise scenarios.

5. Conclusion:

This study evaluated whether retrieval-augmented generation (RAG) improves Transformer robustness for toxic comment detection under noisy text. Based on 30,000 samples with extremely unbalanced data, nested cross-validation of 10 outside and 3 inside, and controllable perturbation, using nested 10*3 cross-validation and controlled perturbations on outer test folds, the study compared Logistic Regression, BERT, and BERT-RAG model performance. Results show that RAG slightly outperforms BERT on clean data (Macro-F1 = 0.905 vs 0.893) but offers limited robustness gains under heavy noise (ΔF1 difference < 0.01). The root cause lies in the fact that noise causes the retrieval embedding to shift and amplify the majority class dominance under a 9-to-1 distribution, thereby weakening the effective evidence for retrieval enhancement from the source. Under the constraints of computing power and time, we fixed K at 32, did not adopt reordering and adaptive gating, nor did we do more random seed and broader search. This might have underestimated the potential upper limit after the robustness modification. Overall, RAG can enhance the accuracy of clean text, but it only shows marginal robustness in noisy and unbalanced Settings. Given that the gains are mainly reflected in clean sets, the future optimization direction tends to start from low-cost renovations: incorporating similarity gating and dynamic fusion, and then introducing rearrangement models based on resources. Contrastive learning is also considered to enhance the anti-noise ability of the retrieval encoder in high-noise and severely unbalanced scenarios.

6. Reference

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